SUMMER INDUSTRIAL INTERNSHIP (AM5200)

Machine Learning Based Classification of Raman Spectra for Oral Cancer Diagnosis

Bidipta Rana
AM25M021 (Mtech 2nd year)
Indian Institute of Technology, Madras | Bhabha Atomic Research Centre, Mumbai

Guide: Dr. R.S. Sengar | Head, Medical Robotics Section, DRHR, BARC

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BACKGROUND

- Oral cancer is a major public health issue in India
- High prevalence due to tobacco use
- Limited diagnostic expertise & lack of mass screening tools
- 60% of oral cancers in developing countries are detected at advanced stages
- Late detection → poor treatment outcomes and considerable costs to patients who cannot afford healthcare

- Early diagnosis improves:
  - Cure rates
  - Reduces treatment cost
  - Lowers morbidity

RAMAN SPECTROSCOPY

What is Raman Spectroscopy?

- A spectroscopic technique used to determine vibrational modes of molecules.
- A laser illuminates the sample; the scattered light is shifted according to molecular vibrational energy levels.

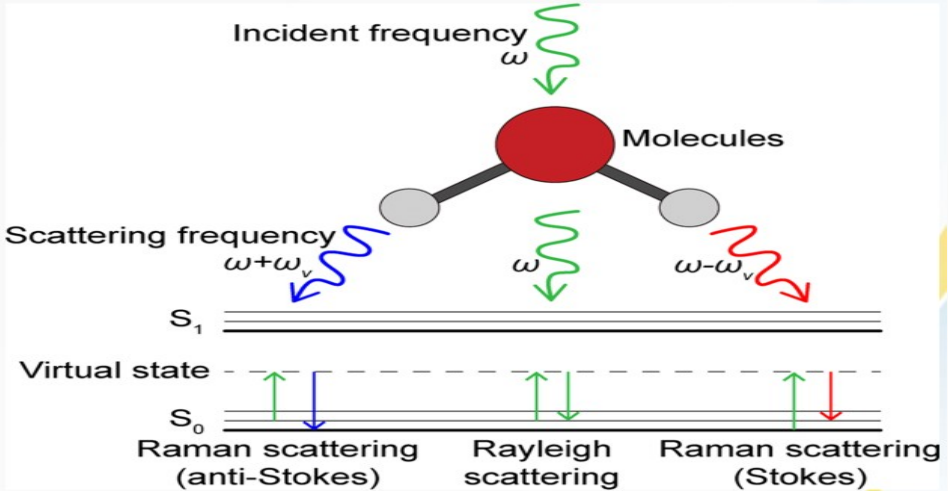
Principle

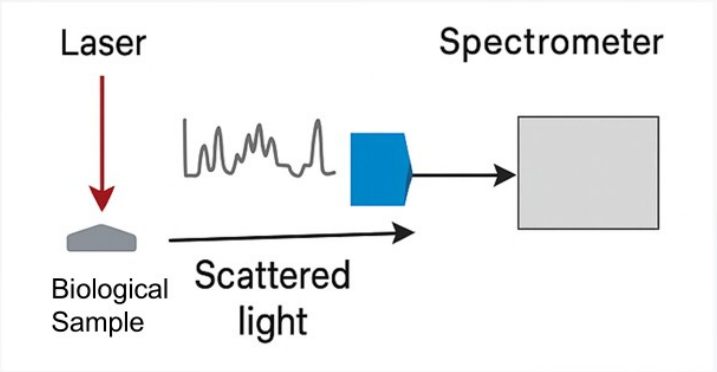
Inelastic scattering of monochromatic light (laser) by molecules produces a wavelength/frequency shift that provides a unique molecular "fingerprint."

$$\Delta\bar{\nu} = \left(\frac{1}{\lambda_0} - \frac{1}{\lambda_s}\right) \times 10^7$$ (Raman shift, cm^{-1}, laser-independent)

Stokes and Anti-Stokes Shifts

- Stokes shift: the molecule gains vibrational energy, so the scattered photon loses the same amount.
- Anti-Stokes shift: the molecule loses vibrational energy, so the scattered photon gains that energy.





Key Features

- Highly specific for detecting molecular bonds and structures
- Non-invasive and label-free (no dyes or contrast agents required)
- Minimal interference from water content
- Healthy tissues contain higher collagen and lipid content
- Cancer cells show changes in nucleic-acid and protein signatures due to rapid cell division

Practical Challenges in Biological Tissue

- **Weak scattering cross-section** — Low intensity of peaks
- **Noisy signal** — Can bury weak Raman peaks
- **Sample heterogeneity** — Inter- and intra-patient variability

Research Gap

- Several laboratory based studies have demonstrated Raman spectroscopy's ability to differentiate between healthy and cancerous oral tissues.
- Existing Raman spectroscopy setups are often bulky, laboratory-based and unsuitable for mass screening
- Biological tissues generate autofluorescence and strong noise background, which obscures weak Raman signals
- Limited availability of high quality in vivo Raman spectra of oral cancers restricts the development of robust AI-based models
- Further research is needed to optimise the acquisition times and resolution of Raman system for clinical applications.

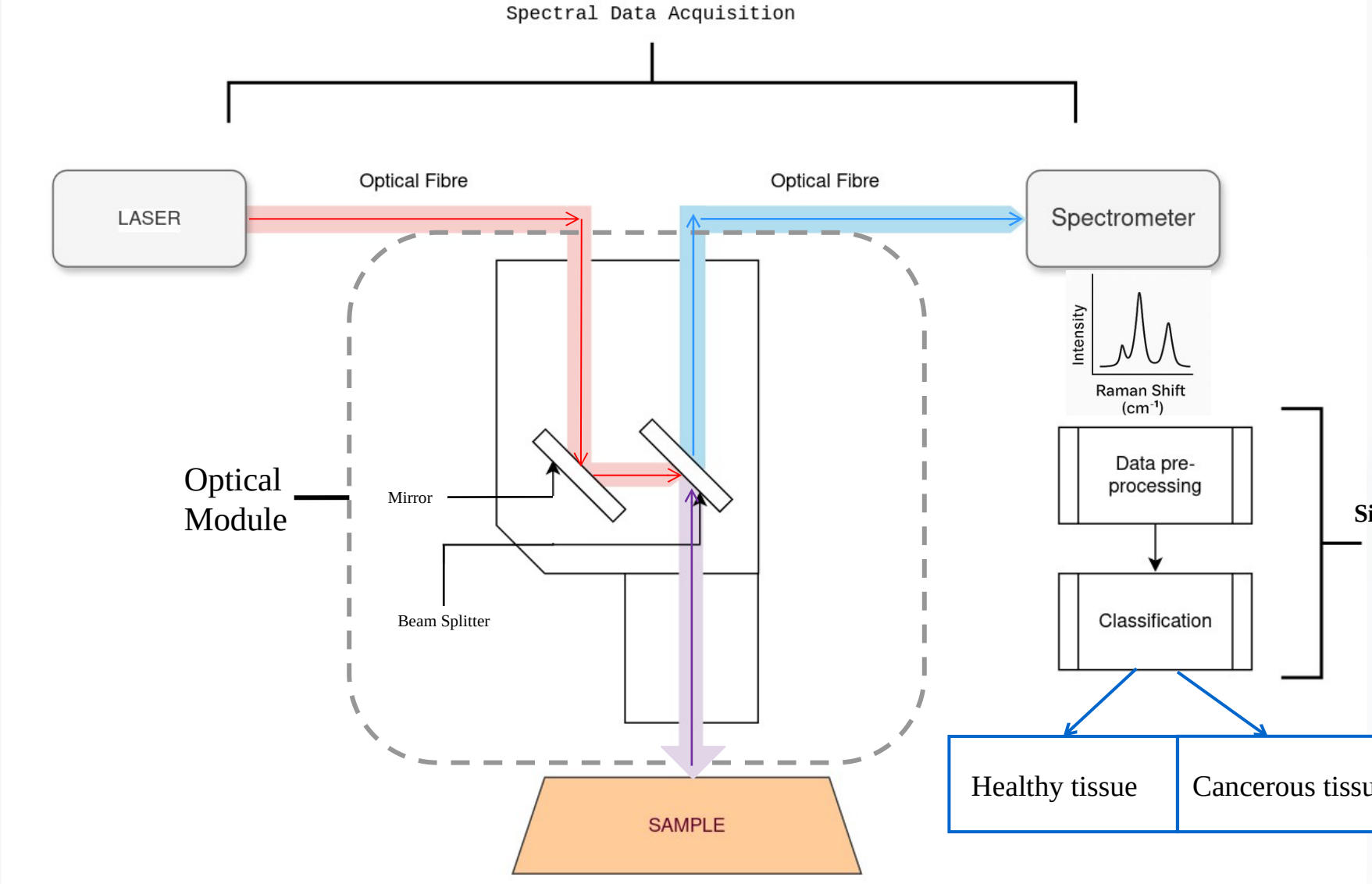
The majority of documented Raman systems remain research-grade prototypes rather than clinically validated tools

The system is currently deployed at ACTREC, Kharghar to collect the patient data

OBJECTIVES OF WORK

- Comparative analysis of various signal pre-processing techniques
- Comparing ML models for real-time classification of cancerous and healthy tissue signals

BLOCK DIAGRAM OF OVERALL RAMAN SYSTEM



PORTABLE SETUP

- It is a portable system, faster to setup and easier to use
- Allows real time screening of cancer detection
- Unlike traditional techniques like biopsies, this system provides instant results
- No need for sample preparation, measurements can be done immediately
- Reduction in diagnostic delays
- Scalable screening tool in rural/urban clinics



Spectral Dataset

345 patients : tobacco-associated, biopsy-verified squamous cell carcinoma
1682 spectra total : ~2 healthy + ~3 tumor spectra per patient (buccal mucosa & tongue)

Male-to-Female Ratio – 9 : 1
Average age of patients – 49 years

Parameter	Value
Excitation wavelength	785 nm
Laser power at tissue	50 mW (spot ⌀ 2 mm)
Integration time	3 s
Irradiance	≈ 1.6 W·cm^{-2} (below ANSI MPE)
Spectral resolution	10 cm^{-1} (830 gr/mm grating)
Fingerprint region used	500 – 2000 cm^{-1}

BASELINE CORRECTION

Asymmetric least squares (AsLS) estimates the baseline by solving the optimization problem:

$$\min_b \left\{ \sum_{i=1}^{N} w_i (y_i - b_i)^2 + \lambda \sum_{i=2}^{N} (\Delta^2 b_i)^2 \right\}$$

The weights are updated iteratively as:

$$w_i = \begin{cases} p, & y_i > b_i \\ 1-p, & y_i \le b_i \end{cases}$$

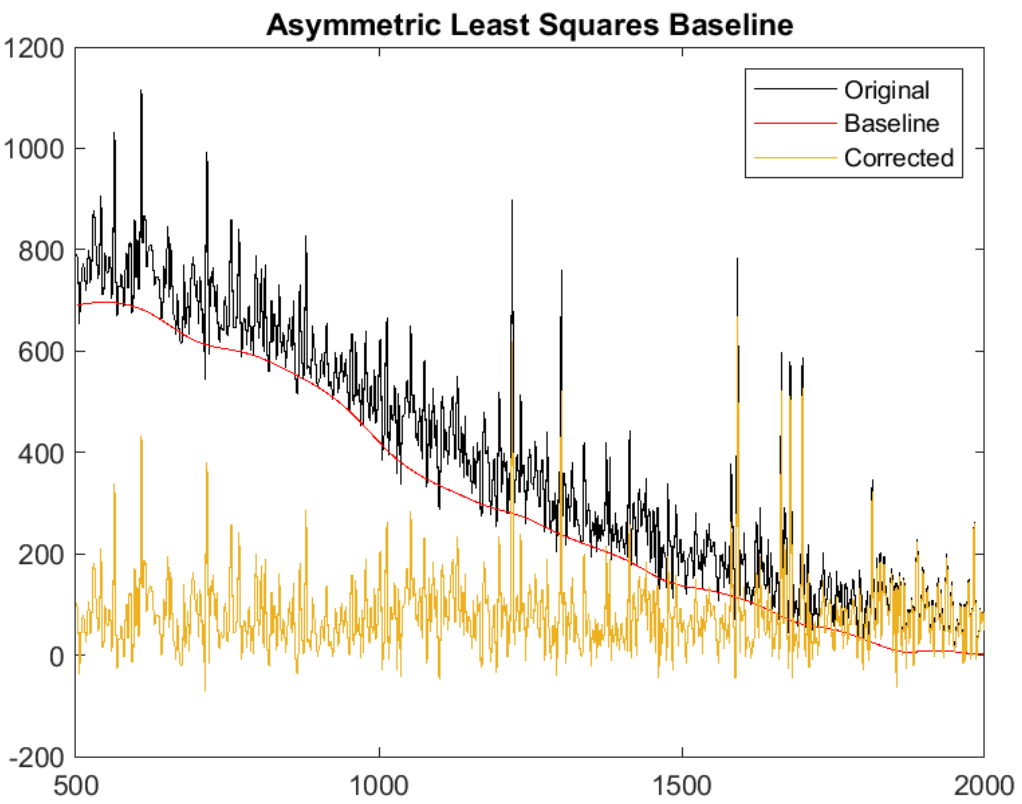
y_i is the measured spectrum,

b_i is the estimated baseline,

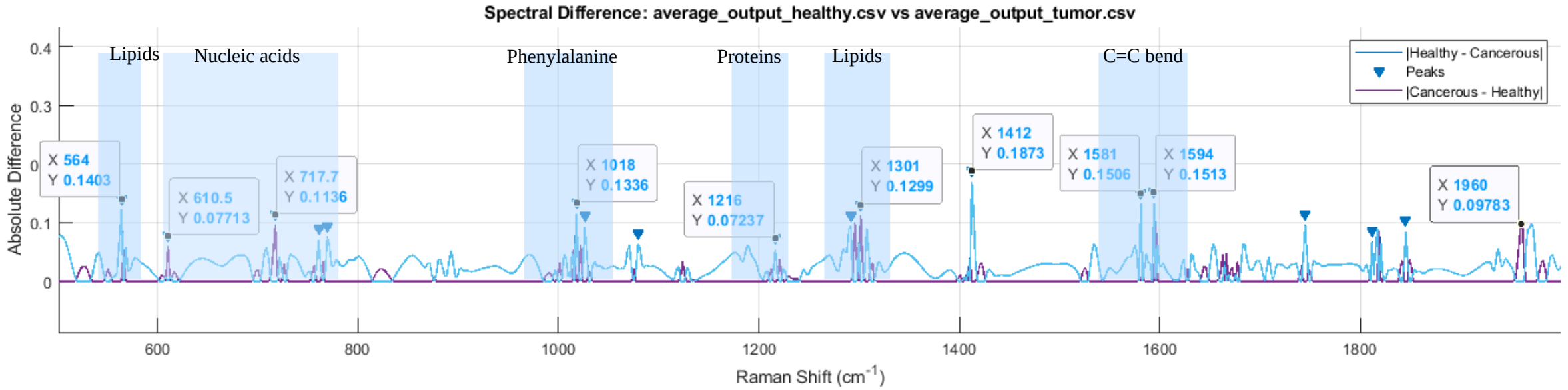
w_i are asymmetric weights,

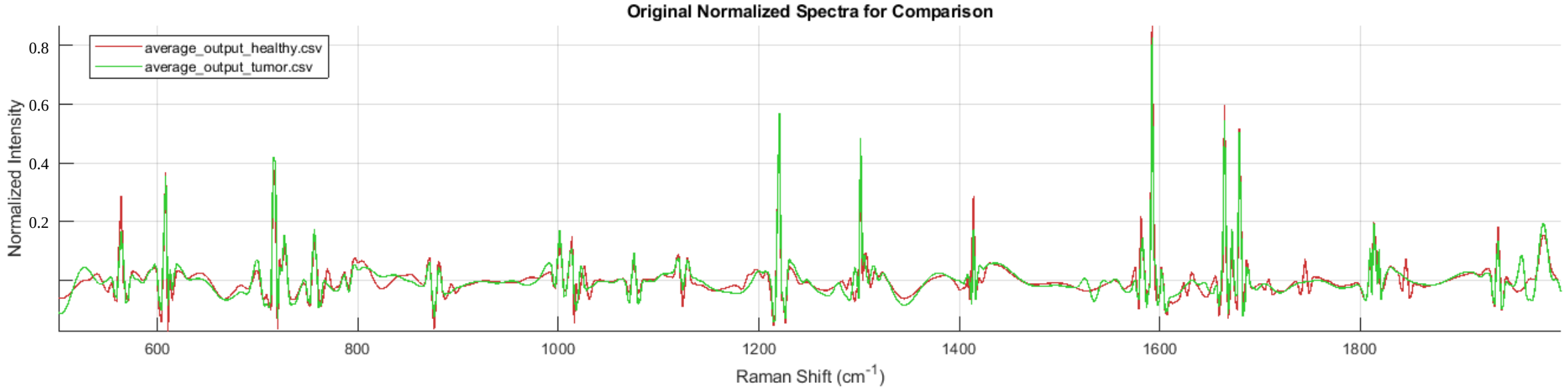
λ is the smoothness parameter controlling baseline curvature,

$\Delta^2 b_i$ is the second-order difference enforcing smoothness.



Difference between Healthy and Cancerous Tissue of a Patient's spectrum



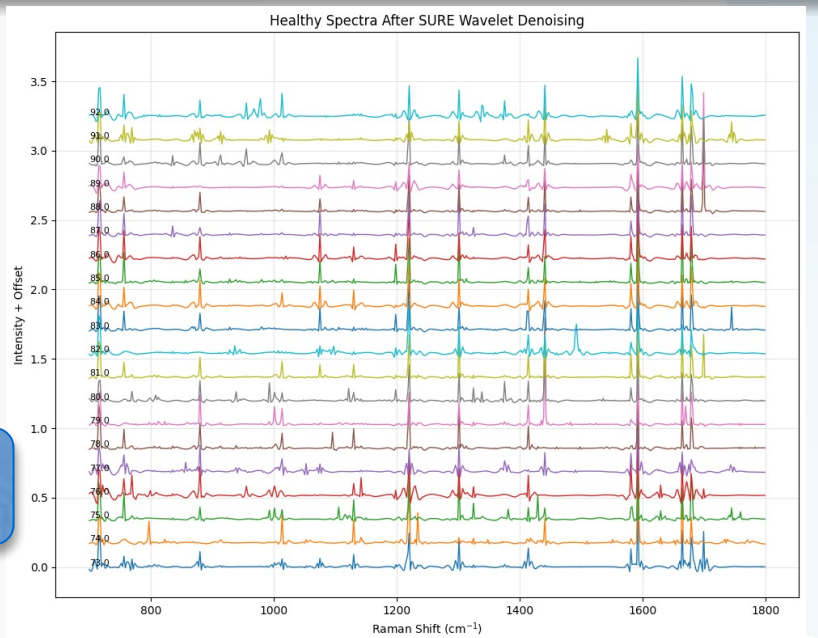


SURE THRESHOLDING (Stein's Unbiased Risk Estimate)

- ***SURE Thresholding (Stein's Unbiased Risk Estimate) :*** It minimizes an estimate of MSE of the denoising without knowing the true signal. The optimal threshold is obtained by minimizing the SURE function:

$$\mathrm{SURE}(T) = N\sigma^2 + \sum \min\left(w_k^2, T^2\right) - 2\sigma^2 \sum_k 1_{\{|w_k| \le T\}}.$$

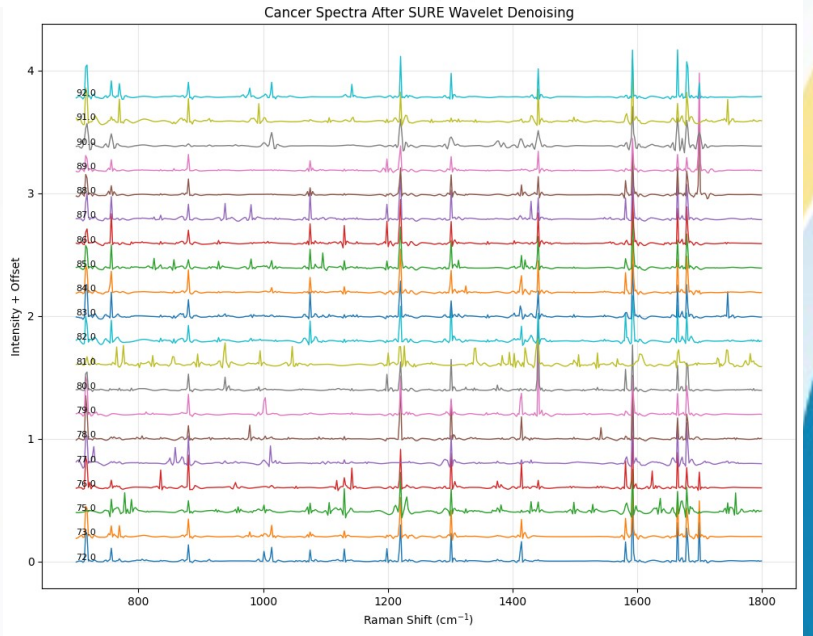
$$T_{\mathrm{SURE}} = \arg\min_T \mathrm{SURE}(T).$$



Feature Extraction

Features with p < 0.05 and effective size > 0.1 were selected, ensuring both statistical significance and practical relevance.

Index	Raman Shift (cm^{-1})	P-Value	Effect Size
20	743.649	0.0061	0.5260
18	739.330	0.0085	0.5037
251	1222.393	0.0107	0.4899
261	1242.257	0.0113	0.4850
17	737.169	0.0118	0.4816
404	1518.990	0.0131	0.4745
530	1752.013	0.0166	0.4593
452	1608.917	0.0197	0.4461
549	1786.315	0.0200	0.4445
431	1569.752	0.0218	0.4381
299	1317.117	0.0231	0.4333
337	1391.009	0.0241	0.4305
455	1614.489	0.0249	0.4282
77	865.442	0.0261	0.4269
548	1784.515	0.0294	0.4155
292	1303.400	0.0305	0.4126
350	1416.069	0.0310	0.4109
555	1797.102	0.0315	0.4115
290	1299.475	0.0325	0.4074
556	1798.898	0.0340	0.4051



Dimensionality Reduction

Why dimensionality reduction?

Raman spectra contain many correlated spectral variables. Dimensionality reduction converts selected spectral features into a small number of latent components while retaining diagnostically relevant information.

Workflow used

761 spectral variables (700–1800 cm^{-1})

31 significant features (t-test + effect size)

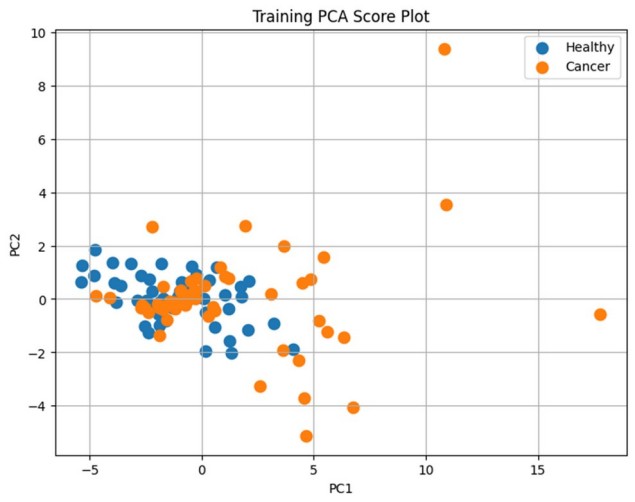
5 components retained for PCA / PLS

Key distinction

PCA Unsupervised	PLS Supervised
Maximizes variance	**Maximizes covariance with class labels**

Training Score Plots

PCA score plot



PCA: 5 PCs retained
≈60% of total variance

PLS score plot



PLS: 5 latent variables
X–Y score correlation = 0.95

Comparison of Different Machine Learning Classification Models

Evaluation Setup

Seven models were evaluated:

- SVM with RBF kernel
- XGBoost
- Random Forest
- Quadratic Discriminant Analysis (QDA)

Each classifier was combined with either PCA or PLS.

A 5-fold cross-validation strategy was used during hyperparameter optimization.

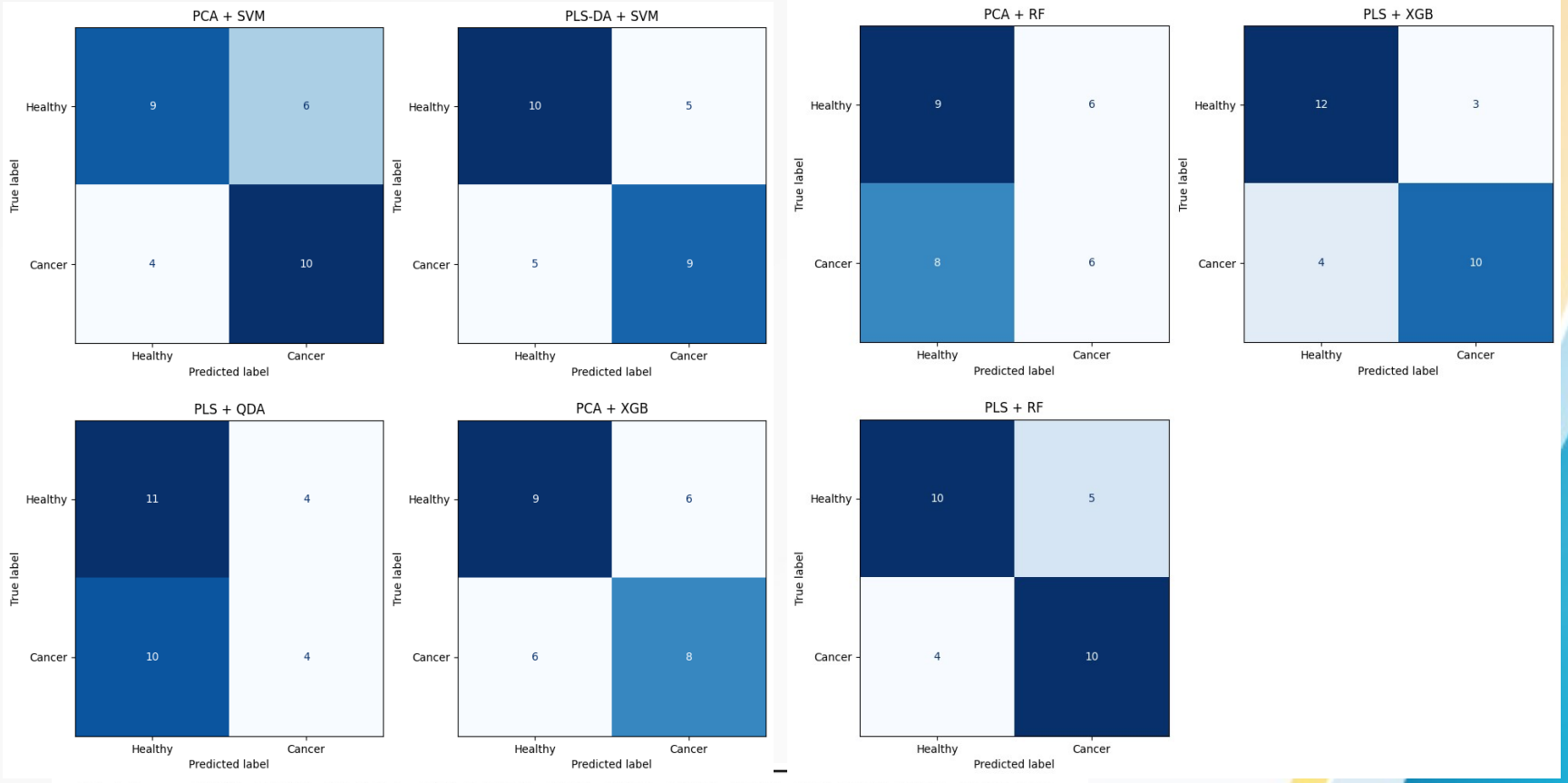
Key Finding: PLS + XGBoost

0.724

Accuracy

PLS + XGBoost achieved the strongest overall metric profile, with precision, recall, specificity, and F1-score all reported at 0.733, 0.733, 0.714, and 0.733 respectively.

Performance Metrics Across Models



Metric	PCA+SVM	PLS-DA+SVM	PLS+QDA	PCA+XGB	PCA+RF	PLS+XGB	PLS+RF
Accuracy	0.655	0.655	0.517	0.586	0.517	**0.724**	0.690
Precision	0.625	0.643	0.500	0.571	0.500	**0.733**	0.667
Recall	0.714	0.643	0.286	0.571	0.429	**0.733**	0.714
Specificity	0.600	0.667	0.733	0.600	0.600	**0.714**	0.667
F1-score	0.667	0.643	0.364	0.571	0.462	**0.733**	0.690
ROC-AUC	0.662	0.762	0.555	0.548	0.533	0.721	0.629

CONCLUSION

This study developed a comprehensive machine learning-based framework for classifying Raman spectra for oral cancer diagnosis. The key findings are:

1. **Preprocessing**: The combination of spectral cropping, ASLS baseline correction, vector normaliza tion, and SURE wavelet denoising effectively improved spectral quality by removing fluorescence background and reducing noise while preserving important spectral features.
2. **Feature Extraction**: Statistical feature selection using t-test and effect size identified 31 significant spectral features that show statistically significant differences between healthy and cancer tissues.
3. **Dimensionality Reduction**: Both PCA and PLS effectively reduced feature dimensionality while preserving discriminatory information. PLS showed better performance when combined with XG Boost.
4. **Model Performance**: The PLS + XGBoost model achieved the best overall performance with 72.4% accuracy, 73.3% sensitivity, and 71.4% specificity.

THANK YOU